=== UNIT 4: SEMICONDUCTOR DIODES, TRANSISTORS & DIGITAL LOGIC ===

>> 4.1 P-N Junction Diodes and Zener Voltage Regulators

Semiconductor physics enables active electronic amplification and digital computation.
* P-N Junction Diode: Formed by joining p-type (acceptor-doped, majority holes) and n-type (donor-doped, majority electrons) semiconductor regions. Majority carriers diffuse across the metallurgical junction, leaving uncompensated ionized dopants forming a space-charge Depletion Region. This creates a built-in barrier potential V_bi (0.7V for Silicon, 0.3V for Germanium).
* Biasing: Forward bias (positive terminal to p-type) narrows depletion width, enabling exponential diffusion current: I = I_0 * [e^{V / (eta * V_T)} - 1]. Reverse bias widens depletion width, reducing current to tiny reverse saturation current I_0.
* Zener Diode: Heavily doped diode exhibiting sharp reverse breakdown. Breakdown mechanisms: (1) Zener breakdown (dominant < 5V: high electric field rips covalent bonds), (2) Avalanche breakdown (dominant > 6V: minority carriers accelerate, impact-ionizing lattice atoms). Used in reverse bias as a precision shunt voltage regulator maintaining constant terminal voltage V_z despite variations in supply voltage or load current.

>> 4.2 Bipolar Junction Transistors (BJT)

* Structure and Doping: Three regions: Emitter (heavily doped to inject carriers), Base (ultra-thin and lightly doped to minimize recombination), and Collector (moderately doped, largest physical area for heat dissipation).
* Active Mode Biasing: Emitter-Base Junction (EBJ) is forward-biased; Collector-Base Junction (CBJ) is reverse-biased. Current relation: I_E = I_B + I_C.
* Common-Emitter (CE) Configuration: Most widely used amplifier configuration offering high voltage gain and high current gain:
- Current Gain Beta (h_fe): Beta = I_C / I_B. Relation to alpha (CB gain): Beta = alpha / (1 - alpha); alpha = Beta / (Beta + 1). Typical Beta = 50 - 300.
- Operating Regions: Active (EBJ forward, CBJ reverse: linear amplifier), Saturation (both forward: closed switch, V_CE,sat ~ 0.2V), Cut-off (both reverse: open switch, I_C ~ 0).

>> 4.3 Digital Electronics and Boolean Logic Gates

* Logic Gates: AND (Y = A . B), OR (Y = A + B), NOT (Y = A'), NAND (Y = (A . B)'), NOR (Y = (A + B)'), XOR (Y = A (+) B = A'B + AB'), XNOR (Y = AB + A'B').
* Universal Gates: NAND and NOR are universal gates because any combinational Boolean circuit (AND, OR, NOT, XOR) can be constructed exclusively using only NAND or only

NOR gates.
* De Morgan's Theorems: (1) (A + B)' = A' . B'; (2) (A . B)' = A' + B'.
* Half Adder: Adds 2 bits: Sum S = A (+) B; Carry C = A . B.
* Full Adder: Adds 3 bits (A, B, C_in): Sum S = A (+) B (+) C_in; Carry C_out = A.B + B.C_in + A.C_in = A.B + C_in.(A (+) B).

[SOLVED PROBLEM 1: Zener Diode Shunt Voltage Regulator Design]

Problem: A 10V Zener diode is used in a shunt regulator circuit with unregulated DC input voltage V_in varying between 15V and 20V. Load resistor R_L varies from 500 Omega to 1,000 Omega. The minimum Zener knee current is I_z,min = 5 mA. Calculate: (1) Value of series current-limiting resistor R_s, (2) Maximum power dissipated in the Zener diode.
Solution Step 1: Determine Load Currents:
Regulated output voltage V_out = V_z = 10 V.
Minimum load current: I_{L, min} = V_z / R_{L, max} = 10 V / 1000 Omega = 10 mA.
Maximum load current: I_{L, max} = V_z / R_{L, min} = 10 V / 500 Omega = 20 mA.
Solution Step 2: Calculate Series Resistor R_s:
Worst-case condition for maintaining regulation occurs when V_in is at its minimum (15V) and load current is at its maximum (20 mA), requiring Zener to maintain minimum knee current (5 mA):
Total supply current I_{s, min} = I_{L, max} + I_{z, min} = 20 mA + 5 mA = 25 mA = 0.025 A.
R_s = (V_{in, min} - V_z) / I_{s, min} = (15 - 10) / 0.025 = 5 / 0.025 = 200 Omega.
Solution Step 3: Maximum Zener Dissipation:
Occurs when V_in is maximum (20V) and load current is minimum (10 mA):
I_{s, max} = (V_{in, max} - V_z) / R_s = (20 - 10) / 200 = 10 / 200 = 50 mA = 0.050 A.
Maximum Zener Current: I_{z, max} = I_{s, max} - I_{L, min} = 50 mA - 10 mA = 40 mA.
Maximum Zener Power: P_{z, max} = V_z * I_{z, max} = 10 V * 0.040 A = 0.40 Watts = 400 mW.
Safety check: A standard 0.5W (500 mW) Zener diode is safely rated for this design.

[SOLVED PROBLEM 2: BJT Common-Emitter DC Operating Point (Q-Point)]

Problem: In a fixed-bias BJT circuit, V_CC = 12 V, R_B = 470 kOmega, R_C = 2.2 kOmega, and Beta = 100. Assuming a silicon transistor with V_BE = 0.7 V, calculate: (1) Base current I_B, (2) Collector current I_C, (3) Collector-Emitter voltage V_CE, (4) Identify whether the transistor operates in Active or Saturation region.
Solution Step 1: Calculate Base Current I_B:
KVL around base-emitter loop: V_CC - I_B * R_B - V_BE = 0.
I_B = (V_CC - V_BE) / R_B = (12 - 0.7) / (470 * 10^3) = 11.3 / 470,000 = 2.404 * 10^-5 A = 24.04 microA.

Solution Step 2: Calculate Collector Current I_C:
Assuming active mode: I_C = Beta * I_B = 100 * 24.04 microA = 2.404 mA.
Solution Step 3: Calculate Collector-Emitter Voltage V_CE:
KVL around collector-emitter loop: V_CC - I_C * R_C - V_CE = 0.
V_CE = V_CC - I_C * R_C = 12 - (2.404 * 10^-3 * 2.2 * 10^3) = 12 - 5.289 = 6.711 Volts.
Solution Step 4: Region Evaluation:
Since V_CE = 6.71 V > V_CE,sat (0.2 V) and CBJ is reverse-biased (V_CB = V_CE - V_BE = 6.71 - 0.7 = 6.01 V > 0), the transistor is operating strictly in the ACTIVE REGION.
Q-Point: (V_CE, I_C) = (6.71 V, 2.40 mA).

[SOLVED PROBLEM 3: Boolean Function Minimization and NAND Implementation]

Problem: Simplify the Boolean expression F(A, B, C) = sum m(1, 3, 5, 7) + d(2) using Karnaugh Map (K-Map) and implement the minimal expression using only 2-input NAND gates.
Solution Step 1: Plot 3-variable K-Map (Variables A on rows, BC on columns):
BC=00 BC=01 BC=11 BC=10
A = 0 0 1(m1) 1(m3) X(d2)
A = 1 0 1(m5) 1(m7) 0
Solution Step 2: Group adjacent 1s and don't-cares:
Minterms m1, m3, m5, m7 form a single 4-cell quad covering column BC = 01 and BC = 11.
In this quad: A changes (0 to 1, eliminated); B changes (0 to 1, eliminated); C remains constant at 1.
Minimal Expression: F(A, B, C) = C.
Alternative Expression with m1, m2, m3, m5: F = A'B + C.
For F = A'B + C: Apply double negation for NAND implementation:
F = ((A'B + C)')' = ((A'B)' . C')'.
Implementation requires 3 NAND gates: (1) Invert A to A', (2) NAND(A', B), (3) Invert C to C', and (4) NAND the results.

[SOLVED PROBLEM 4: Half Adder Implementation via Universal NOR Gates]

Problem: Design a Half Adder (Sum S = A (+) B, Carry C = A . B) exclusively using only 2-input NOR gates. Determine the minimum gate count required.
Solution Step 1: Express Carry C = A . B in NOR form:
Apply De Morgan's theorem: A . B = ((A . B)')' = (A' + B')'.
NOR Formulation of Carry: Invert A and B using NOR gates (A' = A NOR A, B' = B NOR B), then NOR the inverted inputs: C = (A' NOR B') = (A' + B')' = A . B.
Solution Step 2: Express Sum S = A'B + AB' in NOR form:
S = (A + B) . (A . B)' = (A + B) . (A' + B').
Using standard 5-NOR gate universal XOR topology:
Gate 1: N1 = (A NOR B) = (A + B)'.
Gate 2: N2 = (A NOR N1) = (A + (A + B)')' = A' . (A + B) = A'B.

Gate 3: N3 = (B NOR N1) = (B + (A + B)')' = B' . (A + B) = AB'.
Gate 4: Sum S = (N2 NOR N3) = (A'B + AB')' ... invert to get S = A (+) B.
Total Count: A complete half-adder can be implemented with exactly 5 universal NOR gates.

>> 4.4 University Review & Examination Questions

* Q1: Contrast the V-I characteristics and underlying breakdown mechanisms of Zener Breakdown vs. Avalanche Breakdown.
Answer: Zener breakdown occurs in heavily doped p-n junctions with very narrow depletion regions (<10 nm). At low reverse voltages (<5V), the intense electric field (>10^6 V/m) directly ruptures covalent bonds, pulling electrons from the valence band to the conduction band (quantum mechanical tunneling). It exhibits a negative temperature coefficient (breakdown voltage decreases as temperature increases). Avalanche breakdown occurs in lightly doped junctions with wide depletion regions at higher reverse voltages (>6V). Thermally generated minority carriers accelerate across the depletion field, colliding with crystal lattice atoms and knocking out valence electrons (impact ionization), triggering an exponential carrier avalanche. It exhibits a positive temperature coefficient (breakdown voltage increases with temperature due to increased lattice vibrations scattering carriers).
* Q2: Explain the input and output characteristics of a BJT in Common-Emitter (CE) configuration, defining the Early Effect.
Answer: Input characteristics plot I_B vs. V_BE at constant V_CE, resembling a forward-biased diode with cut-in voltage ~0.7V for silicon. Output characteristics plot I_C vs. V_CE at constant I_B, showing three regions: Saturation (V_CE < 0.2V, current rises steeply), Active (nearly horizontal curves where I_C = Beta*I_B), and Cut-off (I_B = 0, tiny leakage current). The Early Effect (Base-Width Modulation) refers to the narrowing of the effective metallurgical base width as reverse-bias collector voltage V_CE increases. A narrower base reduces recombination, causing a slight upward tilt in the active-region output characteristics. When extrapolated backward, all output characteristic curves intersect at a single negative voltage on the V_CE axis termed the Early Voltage (V_A ~ 50-100V).

>> 4.5 Key Takeaways and Revision Points

* P-N diode forward current is exponential; Zener diodes operate in reverse breakdown for voltage regulation.
* BJT CE configuration provides high current gain Beta = I_C / I_B and active operating point (V_CE, I_C).
* NAND and NOR are universal gates; De Morgan's laws: (A+B)' = A'B' and (AB)' = A'+B'.
* Full adder computes Sum = A (+) B (+) C_in and Carry = AB + C_in(A (+) B).